

Meet Vachhani

Surat, Gujarat

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SUMMARY

Integrated M.Tech Artificial Intelligence student with experience in computer vision, image processing, and deep learning. Proficient in developing custom image analysis pipelines, training CNN and Vision Transformer-based models, and building real-time computer vision applications for image classification, facial analysis, and assistive AI solutions.

EDUCATION

Vellore Institute of Technology

2022 – 2027 (Expected)

Integrated M.Tech in Artificial Intelligence

CGPA: 8.12

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, ANN, Data Structures & Algorithms, Probability & Statistics, Linear Algebra, DBMS

TECHNICAL SKILLS

Languages: Python, SQL

Computer Vision: OpenCV, Image Processing, Object Detection, Face Recognition, Image Classification, Vision Transformers (ViT), YOLO

Deep Learning: PyTorch, TensorFlow, CNNs, Transfer Learning, Scikit-learn

AI Frameworks: Hugging Face Transformers, LangChain, LangGraph, FAISS, RAG

Data Processing: Pandas, NumPy, Matplotlib, PySpark, Tableau

Tools & Platforms: Git, GitHub, Linux, Docker, AWS, Jenkins

PROJECTS

PTSD Micro-Expression Detection | 🔗 Link

Nov 2025

Python, TensorFlow, OpenCV, VGG19, Transfer Learning, Image Classification

- Unified CASME, CASME II, SAMM, and SMIC datasets through preprocessing, normalization, and label standardization to create a consistent training corpus.
- Designed a transfer learning framework using VGG19 to identify subtle involuntary facial micro-expressions through discriminative feature extraction.
- Achieved approximately 63% classification accuracy across combined benchmark datasets, demonstrating robust detection of low-intensity facial expressions.

Hybrid CNN-ViTNet for Multi-Class Satellite Image Classification | 🔗 Link

May 2024

PyTorch, Vision Transformer (ViT), CNN

- Proposed a hybrid CNN-ViT architecture combining convolutional feature extraction with transformer-based global context for land-use image analysis.
- Established an end-to-end workflow incorporating preprocessing, augmentation, optimization, validation, and comparative model evaluation.
- Enhanced discrimination of complex satellite scenes by capturing both fine-grained textures and long-range spatial dependencies.

Smart Vision Glasses for the Visually Impaired | 🔗 Link

Jan 2025

YOLOv8, EasyOCR, Google Maps API, Speech Recognition, Text-to-Speech

- Engineered a real-time assistive vision system integrating object detection, face identification, scene understanding, OCR-based text reading, and obstacle awareness from a continuous camera feed.
- Integrated Google Maps navigation with voice-guided directions, enabling independent mobility through speech interaction and contextual audio feedback.
- Optimized concurrent execution of multiple vision components to deliver low-latency environmental awareness for wearable assistive applications.

Bidirectional Sign Language Converter | 🔗 Link

Aug 2025

OpenCV, MediaPipe, Gesture Recognition, SpeechToText

- Developed a bidirectional communication system that converts sign language gestures into text or speech while generating corresponding sign representations from spoken language.
- Applied hand landmark tracking, gesture interpretation, and sequential prediction to accurately understand dynamic signing patterns.
- Facilitated accessible communication between hearing-impaired and non-sign language users through an end-to-end vision-based interaction system.

LEADERSHIP & EXTRACURRICULAR

Operational Manager – AI Club, VIT

May 2025 – Apr 2026

- Oversaw club operations, coordinated cross-functional teams, and executed workshops, competitions, and technical initiatives.

Smart India Hackathon (SIH) – Internal Round Qualifier

Aug 2024 – Nov 2024

- Qualified through the internal selection round among 500+ competing teams at VIT.